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import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
import seaborn as sns
import matplotlib.pyplot as plt
df = sns.load_dataset('titanic')
print(df.head())
print("\n===== Dataset Information =====")
print(df.info())
print("\n===== Statistical Summary =====")
print(df.describe())
print("\n===== Missing Values =====")
print(df.isnull().sum())
df['age'] = df['age'].fillna(df['age'].median())
df['embark_town'] = df['embark_town'].fillna(
    df['embark_town'].mode()[0])
)
df.drop(columns=['deck'], inplace=True)
le = LabelEncoder()
df['sex'] = le.fit_transform(df['sex'])
df['embark_town'] = le.fit_transform(df['embark_town'])
df.drop(
    columns=['embarked', 'class', 'who', 'alive',
            'adult_male', 'alone'],
    inplace=True
)
scaler = StandardScaler()
numerical_cols = ['age', 'fare']
df[numerical_cols] = scaler.fit_transform(
    df[numerical_cols]
)
X = df.drop('survived', axis=1)
y = df['survived']

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
print("\nTraining Data Shape:", X_train.shape)
print("Test Data Shape:", X_test.shape)

print("\nFirst 5 Rows of Training Data:")
print(X_train.head())

```

0	survived	891 non-null	int64
1	pclass	891 non-null	int64
2	sex	891 non-null	object
3	age	714 non-null	float64
4	sibsp	891 non-null	int64
5	parch	891 non-null	int64
6	fare	891 non-null	float64
7	embarked	889 non-null	object
8	class	891 non-null	category

```
===== Missing Values =====
```

```
survived      0
pclass        0
sex           0
age          177
sibsp         0
parch         0
fare          0
embarked      2
class         0
who           0
adult_male    0
deck         688
embark_town   2
alive         0
alone         0
dtype: int64
```

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Training Data Shape: (712, 7)
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Test Data Shape: (179, 7)
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First 5 Rows of Training Data:
```

	pclass	sex	age	sibsp	parch	fare	embark_town
331	1	1	1.240235	0	0	-0.074583	2
733	2	1	-0.488887	0	0	-0.386671	2
382	3	1	0.202762	0	0	-0.488854	2
704	3	1	-0.258337	1	0	-0.490280	2
812	2	0	1.705224	1	2	0.010700	2